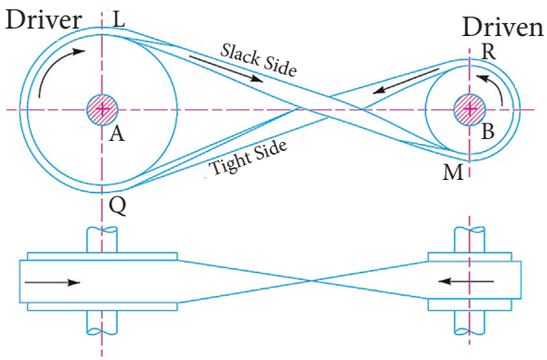
CROSSED BELT DRIVE:

In this type of belt drive, the belt is crossed between the pulleys. The belt connects the top portion of the driver with the lower portion of follower. The grip between the belt and pulley is greater because of the crossed nature of the belt. The pulleys connected by the cross belt arrangement rotate in the opposite directions. If the driver rotates in clockwise direction, the follower will rotate in the anticlockwise direction.

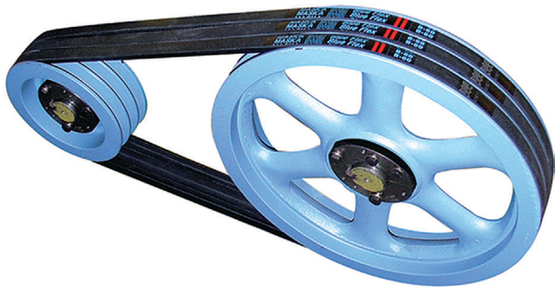


Crossed Belt Drive

TYPES OF BELT

Belt is usually made from leather, rubber and canvas thread in a moulded form. The two ends of a belt are connected by hooks and pins. Generally two forms of belts are used.

1. Flat belt.
2. V – belt



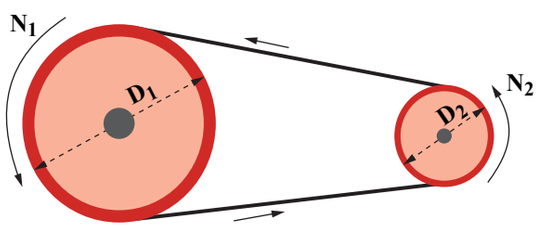
V-BELT

FLAT BELT
1. Cross section thickness 0.75mm to 5mm.
2. Transmission of power is low
3. Flat face pulleys are used
V- BELT
1. Cross section thickness 8mm to19mm.
2. Transmission of power is high
3. V-groove pulleys are used.

VELOCITY RATIO OF A BELT DRIVE:

Velocity ratio of a belt drive is the ratio of number of revolutions of follower to the number of revolutions of driver in a particular time.

If D_1 and D_2 are the diameter of driver and follower and N_1 and N_2 are the number of revolutions per minute of the driver and the follower.



$$\text{Velocity ratio} = \frac{N_2}{N_1} = \frac{D_1}{D_2}$$

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